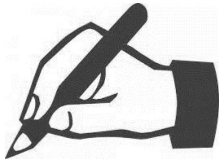

- Q1. Recall the reactivity series in the correct order

- Q2. Write a word equation for the reaction between iron sulfate and magnesium

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DO NOW

- Q1.

Potassium	Most reactive	K
Sodium		Na
Calcium		Ca
Magnesium		Mg
Aluminium		Al
Carbon		C
Zinc		Zn
Iron		Fe
Tin		Sn
Lead		Pb
Hydrogen		H
Copper		Cu
Silver		Ag
Gold		Au
Platinum	Least reactive	Pt

- Q2. iron sulfate + magnesium → magnesium sulfate + iron

3.5

Keywords

Coke
Oxidation
Blast furnace
Reduction

Extraction of metals

Learning Objective:

1. Represent displacement reactions with carbon, metal oxides and iron using formulas and equations.
2. Explain how mass is conserved in the extraction of metals

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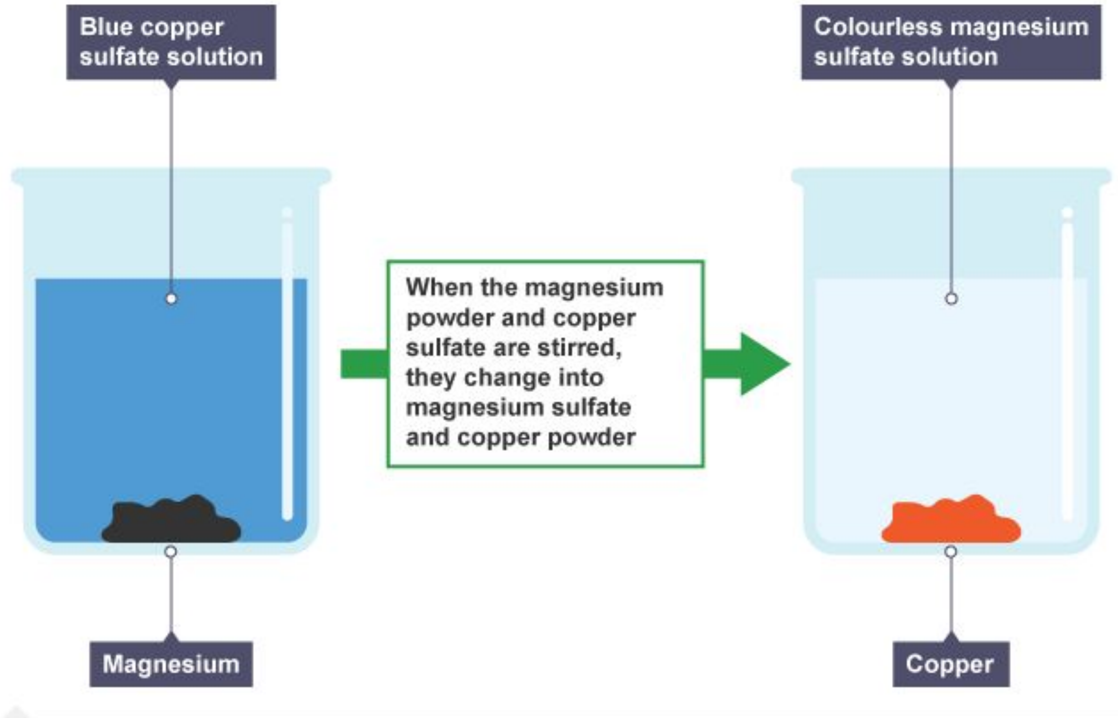
Re-cap on displacement reactions

- Q. What is displacement reaction?

- A chemical reaction in which one substance takes the place of another in a compound.



A displacement reaction

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CONNECTION

Using carbon to extract iron

- Looking at the reactivity series Carbon is more reactive than iron, therefore it can displace iron from iron compounds

iron(III) oxide + carbon → iron + carbon monoxide

Metal	Method	Reactivity
Potassium	Electrolysis of molten compounds	Most reactive
Sodium		
Lithium		
Calcium		
Magnesium		
Aluminium		
(Carbon)		
Zinc	Heating with carbon	
Iron		
Copper		
Gold	Various chemical reactions	Least reactive

Extracting a metal by heating with carbon is cheaper than using electrolysis.

Q. Where can carbon be found?

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ACTIVATION

- Burning wood is mostly carbon, it was thought that the addition of lead and tin ore to wood fires resulted in the formation of lead and tin.

- Ore- A natural rock from which a metal is extracted from

- The ore needs to be chemically separated before the metal can be used as an element.

Coal is a richer source of carbon, when it is burned slowly over 1000◦C, in the absence of oxygen, coke is made.

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ACTIVATION

- Carbon is more reactive than iron, tin, lead and copper, therefore to separate the ore from the metal, carbon- displacement reactions.

Look at the example below, copy out the equation and in a few sentences explain what has happened in this displacement reaction

zinc oxide + carbon → zinc + carbon dioxide

Carbon has removed the oxygen from the metal zinc, this removal of oxygen is called **reduction**.

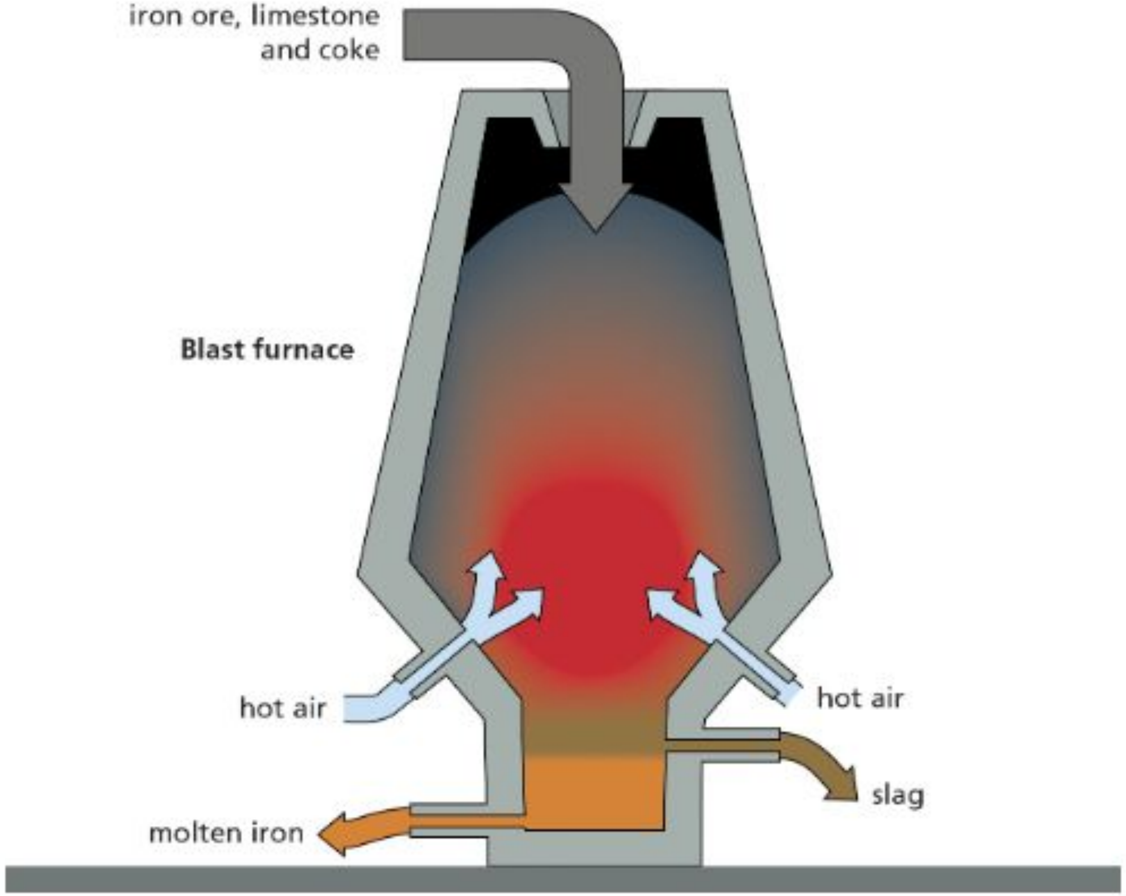
Carbon has gained the oxygen that was removed, this gain of oxygen is called **oxidation**

Extraction of iron

- Iron ore exists in the form of iron oxide (Fe_2O_3)- *iron that has reacted with oxygen.*

- To extract iron from its ore, powdered coke is added to the crushed iron ore along limestone- *which removes impurities.*

- All the ingredients are roasted in a blast furnace



- Air is injected into the bottom of the furnace to allow carbon to act as a reducing agent (remove the oxygen)

- The blast furnace operates at temperatures up to 1650◦C

Stick the following diagram in your book

ACTIVATION

To allow carbon to act as a reducing agent, carbon has to be heated to produce carbon monoxide.

The reducing agent used is carbon monoxide.

Carbon monoxide is made by heating carbon.
In industry the process has many steps but put simply:

carbon + oxygen	→	carbon monoxide	
iron oxide + carbon monoxide	→	iron + carbon dioxide	
$2C + O_2$	→	$2CO$	(i)
$Fe_2O_3 + 3CO$	→	$2Fe + 3CO_2$	(ii)

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ACTIVATION

Key Questions- Worksheet

Decide whether each of these pairs of reactants will react or not.

If they will react, complete the equation; if not, write 'no reaction'.

1. Magnesium + silver nitrate → ______________________________
2. Copper nitrate + zinc → ______________________________
3. Silver nitrate + iron → ______________________________
4. Copper + magnesium sulfate → ______________________________
5. $ZnSO_4$ + Mg → ______________________________
6. Cu + $MgSO_4$ → ______________________________
7. Mg + $FeSO_4$ → ______________________________
8. $CuSO_4$ + Fe → ______________________________
9. Zn + $PbSO_4$ → ______________________________
10. $PbSO_4$ + Fe → ______________________________

Copy out and completed the following displacement reactions, if there is no reaction then write 'No reaction'

Key Questions- Answers

1. Magnesium + silver nitrate → magnesium nitrate + silver
2. Copper nitrate + zinc → zinc nitrate + copper
3. Silver nitrate + iron → iron nitrate + silver
4. Copper + magnesium sulfate → no reaction
5. $ZnSO_4 + Mg \rightarrow MgSO_4 + Zn$
6. $Cu + MgSO_4 \rightarrow$ no reaction
7. $Mg + FeSO_4 \rightarrow MgSO_4 + Fe$
8. $CuSO_4 + Fe \rightarrow FeSO_4 + Cu$
9. $Zn + PbSO_4 \rightarrow Pb + ZnSO_4$
10. $PbSO_4 + Fe \rightarrow Fe\ SO_4 + Pb$

Predicting the mass of iron

- The balanced equation below shows the overall reaction that takes place inside the blast furnace:

- $2FeO_3 + 3C \longrightarrow 3CO_2 + 4Fe$

- By knowing the mass of the reactants (iron ore and carbon) that was added to the furnace and knowing that mass is always conserved, we can predict the amount of the metal iron was made

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Conservation of Mass

The law of conservation of mass states that "no atoms are lost or made during a chemical reaction so the mass of the products equals the mass of the reactants".

$$\underbrace{A + BC}_{\text{Reactants are to the left of the arrow}} \rightarrow \underbrace{AB + C}_{\text{Products are to the right of the arrow}}$$

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Example…

If 100 tonnes of iron ore was added to the furnace and it had a purity where 20 per cent of the mass of the ore was iron, how much iron metal could theoretically be made?

The amount of iron entering the furnace = amount of ore × percentage purity = 100 × 0.2 = 20 tonnes.

Because mass is conserved, the amount going in should equal the amount coming out.

So the theoretical amount of iron metal made would be 20 tonnes.

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Practice questions

5. Magnetite (Fe_3O_4) is another ore of iron used in the blast furnace. Write a balanced equation to show the products made.

6. How much iron metal could be made from 250 tonnes of magnetite with a purity where 30 per cent of its mass is iron?

- Answer in full sentences and show your working out for question 6

Answers

- 5. $Fe_3O_4 + 2C \longrightarrow 2CO_2 + 3Fe$

- 6. 250 x 0.3 = 75 tonnes

Extracting copper, zinc and lead

- Extracting copper:

Malachite ($CuCO_3$) is heated to over 200°C to form copper oxide.

- The following equations demonstrates the process of extracting copper from copper ore:

$$CuCO_3 \rightarrow CuO + CO_2$$

Copper oxide is roasted with carbon to make copper metal.

$$2CuO + C \rightarrow 2Cu + CO_2$$

This is copper metal.

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ACTIVATION

- <u>Extracting lead and zinc</u>

ACTIVATION

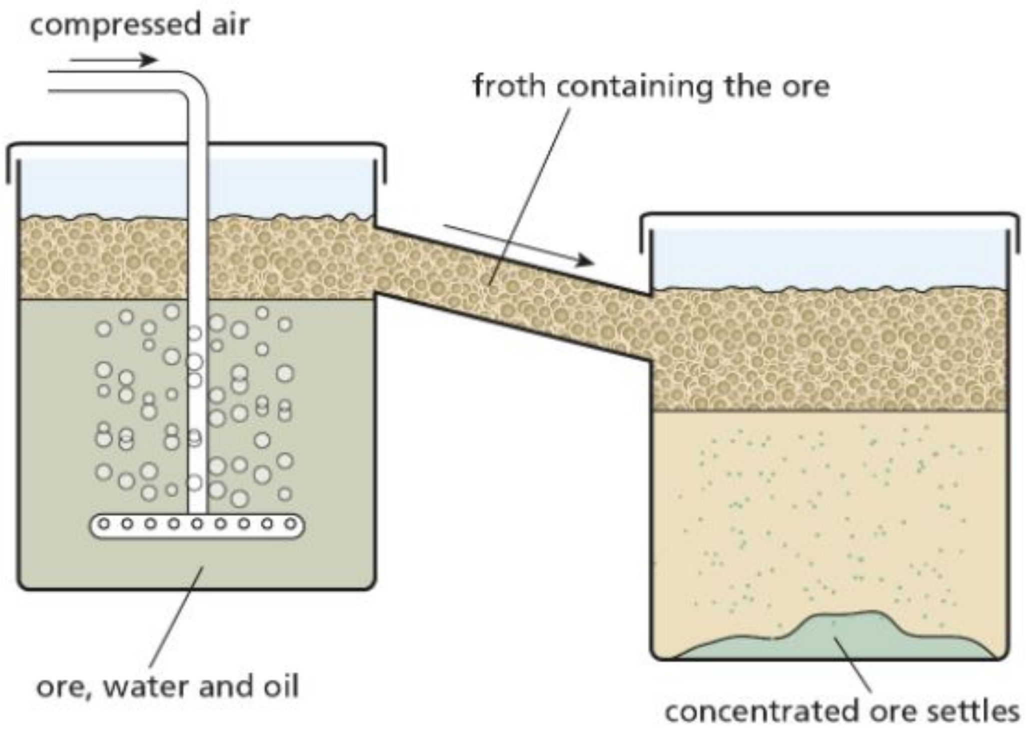
- Galena (lead sulphide) is an example of a lead ore.

- 1. The ore is crushed into small particles

- 2. These are added to a mixture of oil and water

- 3. Air is blown upwards to push the lead ore particles to the top, separating them from other impurities

- 4. This is called 'froth flotation'



Stick the flow chart into your book

- From the information in your flow chart, write down 2 separate word equations for the following reaction.

lead sulfide + oxygen → lead oxide + sulfur dioxide

lead oxide + carbon → lead + carbon dioxide

- Zinc ore (saphalerite) is zinc sulphide, before it can be displaced with carbon, the zinc sulphide needs to be changed to zinc oxide.
- The same process in extracting lead is carried out:

- 1. Blowing hot air through a concentrated mixture of zinc sulphide and water- froth flotation.

↓

- 2. The concentrated zinc sulphide is then roasted in air to make zinc oxide.

↓

- 3. The oxide is then removed by roasting zinc oxide with carbon

Answer the exam questions on your worksheet

The word equation below shows a reaction used in an industrial process.

chromium oxide + aluminium → chromium + aluminium oxide

The reaction is highly exothermic.

(a) What is an exothermic reaction?

____

____

(2)

(b) Name the products of this reaction.

____

(1)

(c) In the reaction one substance is reduced.

(i) Name the substance which is reduced.

____

(1)

(ii) What happens to the substance when it is reduced?

____

____

(1)

(Total 5 marks)

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CONSOLIDATION

Answers

(a)	gives out		
	heat		
	each for 1 mark		2
(b)	chromium and aluminium oxide		1
(c)	(i)	chromium oxide	1
	(ii)	oxygen removed/gains electrons	1
			[5]

Calculating yield

The symbol equations for the production of lead and zinc from their oxides are:

$2PbO + C \rightarrow 2Pb + CO_2$

$2ZnO + C \rightarrow 2Zn + CO_2$

- The process of extracting metals from their ores are inefficient because not all the metal is removed from the ore.

- Therefore the yield of the process is less than 100%

$$\text{yield} = \frac{\text{actual amount produced}}{\text{theoretical amount}} \times 100\%$$

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ACTIVATION

$$\text{yield} = \frac{\text{actual amount produced}}{\text{theoretical amount}} \times 100\%$$

- The following formula is used to help calculate how much of the metal could have been extracted in theory.

- Using the formula calculate the questions below:

5. 500 tonnes of zinc is produced from an ore containing 800 tonnes. Work out the yield of the process.
6. 446 tonnes of lead oxide is roasted with coke, and 200 tonnes of lead are made. What is the yield of the process? Assume that the lead oxide is 50 per cent by mass lead, and use ideas from Topic 3.5 to help you.

Check your answers

5
Yield = 500 ÷ 800 × 100 % = 62.5 %

6	Answers
50% of 446 = 223 tonnes of lead oxide are in the ore; 200 ÷ 223 × 100 % = 90 % yield	